NO ABELIAN $(64681, 441, 3)$-DIFFERENCE SET EXISTS

Abstract. We show that no abelian group of order 64681 admits a $(64681, 441, 3)$-difference set. The proof is short: since $64681 = 71 \cdot 911$ is squarefree, every abelian group of that order is cyclic. The first multiplier theorem makes 73 a multiplier, and after translation any putative difference set is invariant under multiplication by 73 on C_{64681}. Every nonzero 73-orbit has size 35, so an invariant set has size $\delta + 35a$ with $\delta \in \{0, 1\}$. This cannot equal 441.

1. Introduction

Gordon's 2022 analysis of abelian difference sets with small λ leaves $(64681, 441, 3)$ on the residual list of open abelian $\lambda = 3$ cases [1]. We record a direct multiplier-orbit obstruction that removes this tuple.

Theorem 1. *No abelian group of order* 64681 *admits a* $(64681, 441, 3)$-*difference set.*

Proof. Let G be an abelian group of order

$$64681 = 71 \cdot 911.$$

Since the order is squarefree, G is cyclic, so it is enough to rule out a difference set in C_{64681}.

Assume for contradiction that $D \subseteq C_{64681}$ is a $(64681, 441, 3)$-difference set. Its order is

$$n = k - \lambda = 441 - 3 = 438 = 2 \cdot 3 \cdot 73.$$

Because 73 is a prime divisor of n with $73 > \lambda$ and $\gcd(73, 64681) = 1$, the first multiplier theorem implies that 73 is a numerical multiplier of D. Thus there exists $s \in C_{64681}$ such that

$$73D = D + s.$$

Since

$$\gcd(73 - 1, 64681) = \gcd(72, 64681) = 1,$$

we may choose $g \in C_{64681}$ with $(73-1)g = -s$. Then the translate $D' = D+g$ satisfies

$$73D' = 73D + 73g = D + s + 73g = D + g = D'.$$

Replacing D by D', we may assume from the start that $73D = D$.

We now compute the orbit sizes of the map $x \mapsto 73x$ on C_{64681}. The point 0 is fixed. If x is a nonzero multiple of 911, then its orbit size is $\mathrm{ord}_{71}(73)$. Since $73 \equiv 2 \pmod{71}$ and $2^{35} \equiv 1 \pmod{71}$, we have $\mathrm{ord}_{71}(73) = 35$.

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Likewise, if x is a nonzero multiple of 71, then the orbit size is $\mathrm{ord}_{911}(73)$, and a direct computation gives $\mathrm{ord}_{911}(73) = 35$. For a unit x, the orbit size is

$$\mathrm{lcm}\big(\mathrm{ord}_{71}(73), \mathrm{ord}_{911}(73)\big) = 35.$$

Therefore every nonzero orbit has size 35.

Because D is 73-invariant, it is a union of 73-orbits. Hence

$$|D| = \delta + 35a$$

for some integer $a \geq 0$ and some $\delta \in \{0, 1\}$ according to whether $0 \in D$. But $|D| = 441$, and neither 441 nor 440 is divisible by 35. This contradiction proves that no such difference set exists. □

Remark 2. The obstruction is entirely orbit-theoretic: once the valid multiplier 73 is chosen, the orbit-size semigroup on C_{64681} misses the required weight 441.

References

[1] Daniel M. Gordon, *On difference sets with small lambda*, J. Algebraic Combin. **55** (2022).

[2] E. S. Lander, *Symmetric Designs: An Algebraic Approach*, Cambridge University Press, 1983.